

# Cambridge International AS & A Level

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## MATHEMATICS

**9709/11**

Paper 1 Pure Mathematics 1

October/November 2023

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

## INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

## INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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- 1 (a) Expand  $(1 + 3x)^6$  in ascending powers of  $x$  up to, and including, the term in  $x^2$ . [2]

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- (b) Hence find the coefficient of  $x^2$  in the expansion of  $(1 - 7x + x^2)(1 + 3x)^6$ . [2]

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- 2** A line has equation  $y = 2cx + 3$  and a curve has equation  $y = cx^2 + 3x - c$ , where  $c$  is a constant.

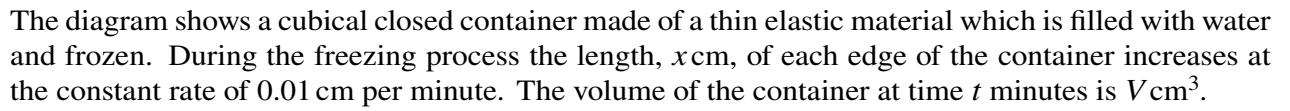
Showing all necessary working, determine which of the following statements is correct.

- A** The line and curve intersect only for a particular set of values of  $c$ .

- B** The line and curve intersect for all values of  $c$ .

- C** The line and curve do not intersect for any values of  $c$ . [4]

[illegible]



[3]

[illegible]

- 4** The transformation R denotes a reflection in the  $x$ -axis and the transformation T denotes a translation of  $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ .

- (a)** Find the equation,  $y = g(x)$ , of the curve with equation  $y = x^2$  after it has been transformed by the sequence of transformations R followed by T. [2]

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- (b)** Find the equation,  $y = h(x)$ , of the curve with equation  $y = x^2$  after it has been transformed by the sequence of transformations T followed by R. [2]

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- (c)** State fully the transformation that maps the curve  $y = g(x)$  onto the curve  $y = h(x)$ . [2]

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- 5 (a) Show that the equation

$$4 \sin x + \frac{5}{\tan x} + \frac{2}{\sin x} = 0$$

may be expressed in the form  $a \cos^2 x + b \cos x + c = 0$ , where  $a$ ,  $b$  and  $c$  are integers to be found. [3]

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- (b) Hence solve the equation  $4 \sin x + \frac{5}{\tan x} + \frac{2}{\sin x} = 0$  for  $0^\circ \leq x \leq 360^\circ$ . [3]

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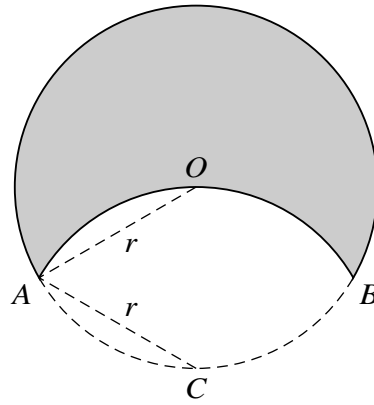
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The diagram shows a motif formed by the major arc  $AB$  of a circle with radius  $r$  and centre  $O$ , and the minor arc  $AOB$  of a circle, also with radius  $r$  but with centre  $C$ . The point  $C$  lies on the circle with centre  $O$ .

- (a) Given that angle  $ACB = k\pi$  radians, state the value of the fraction  $k$ . [1]

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- (b) State the perimeter of the shaded motif in terms of  $\pi$  and  $r$ . [1]

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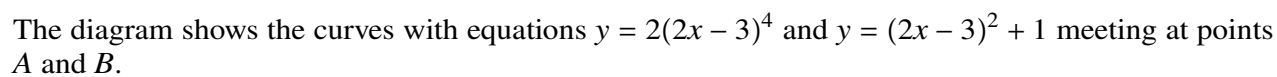


- 7 The sum of the first two terms of a geometric progression is 15 and the sum to infinity is  $\frac{125}{7}$ . The common ratio of the progression is negative.

Find the third term of the progression.

[7]

[illegible]



- [illegible]

(b) Find the exact area of the shaded region.

[5]

[illegible]

- 9 (a) Express  $4x^2 - 12x + 13$  in the form  $(2x + a)^2 + b$ , where  $a$  and  $b$  are constants. [2]

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The function  $f$  is defined by  $f(x) = 4x^2 - 12x + 13$  for  $p < x < q$ , where  $p$  and  $q$  are constants. The function  $g$  is defined by  $g(x) = 3x + 1$  for  $x < 8$ .

- (b) Given that it is possible to form the composite function  $gf$ , find the least possible value of  $p$  and the greatest possible value of  $q$ . [3]

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(c) Find an expression for  $gf(x)$ .

[1]

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The function  $h$  is defined by  $h(x) = 4x^2 - 12x + 13$  for  $x < 0$ .

(d) Find an expression for  $h^{-1}(x)$ .

[3]

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**10** A curve has a stationary point at  $(2, -10)$  and is such that  $\frac{d^2y}{dx^2} = 6x$ .

**(a)** Find  $\frac{dy}{dx}$ . [3]

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**(b)** Find the equation of the curve. [3]

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- (c) Find the coordinates of the other stationary point and determine its nature. [3]

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- (d) Find the equation of the tangent to the curve at the point where the curve crosses the  $y$ -axis. [2]

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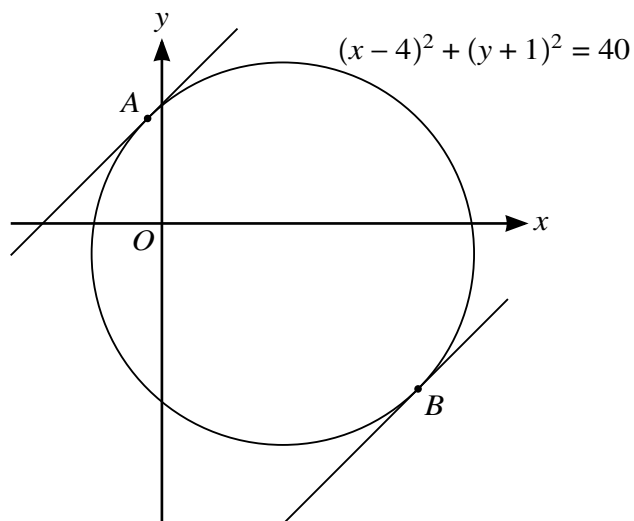
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The diagram shows the circle with equation  $(x - 4)^2 + (y + 1)^2 = 40$ . Parallel tangents, each with gradient 1, touch the circle at points  $A$  and  $B$ .

- (a) Find the equation of the line  $AB$ , giving the answer in the form  $y = mx + c$ . [3]

[illegible]

(b) Find the coordinates of  $A$ , giving each coordinate in surd form.

[4]

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(c) Find the equation of the tangent at  $A$ , giving the answer in the form  $y = mx + c$ , where  $c$  is in surd form.

[2]

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[illegible]

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